

GARCH-LSTM Long/Short Pair Trading Strategies on ETH/BTC pair

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Abstract

Conventional pairs trading strategies rely on historical correlations and assume mean-reversion, which isn't substantial to highly volatile and non-linear dynamic markets such as the cryptocurrency market. The project proposes leveraging Machine learning, particularly combining Generalized Autoregressive Conditional Heteroskedasticity and Long Short Term Memory model with Attention mechanism and an Altcoin Season Index regime-switching filter to pair trading in such a dynamic market. This project includes model development, backtesting and evaluation, aiming to outperform traditional statistical methods in terms of adaptability and profitability.

Keywords: *Pairs trading, LSTM, GARCH, EGARCH, machine learning, financial markets, cryptocurrency, attention mechanisms, ASI*

1. Introduction

In today's fast-evolving financial markets, particularly in the cryptocurrency market, traditional statistical trading strategies often struggle to adapt to financial markets' dynamic and non-linear nature.

This project aims to enhance pair trading strategies by integrating the machine learning technique Generalized Autoregressive Conditional Heteroskedasticity (GARCH) for volatility and Long Short-Term Memory (LSTM) with Attention mechanism for price movement. Using ETH-USD and BTC-USD as the primary asset pair, this model aims to capture time-dependent trends and patterns that conventional statistical methods may overlook. By predicting future spread movements dynamically, the proposed strategy seeks to develop a dynamic trading strategy that better adapts to market fluctuations and improves overall risk-adjusted returns.

2. Literature Review

2.1 Long/Short Pair Trading Strategies and its Limitations

Long/short pair trading is a market-neutral strategy based on the relative price movement between two correlated assets. Historically, pair trading as a traditional statistical arbitrage strategy has been based on cointegration, mean-reversion assumption and linear models like ordinary least squares (OLS). Yet classical models frequently fail in non-stationary contexts, with structural breaks and nonlinear dependencies. This shift to a new paradigm is where in addition to all the existing time series approaches we can consider a bunch of rules that predict the behaviour of a model such as Machine Learning (ML) and Deep Learning (DL), which are capable of learning complex patterns from higher dimensional data and adapt to changes in market regimes nearly real-time.

Standard pair trading strategies are mainly based on stages of identifying cointegrated pairs, then entering the market for a long position or a short position at times when price ratios differ from historical average values. Methods mentioned in Hands-On Financial Trading with Python, by Pik and Ghosh, are based on the assumptions of stationarity and

linearity. While these methods are simple and interpretable, they fail to adapt to dynamic data in noisy environments or when different regimes appear.

2.2 Deep Learning in Financial Trading

As highlighted on Pythonic Quant by Van Der Post, ML goes beyond rule-based systems in finance as it learns from raw information. For example, Random Forest and XGBoost convert critics interpreted feature importance and functioning in heterogeneous data scenarios. Ensemble methods are generally the starting point for improving standard delicacies, partially due to their accuracy with tabular data and their capacity for nonlinearity, as they do not necessitate characteristic engineering.

Kaabar has an exhaustive review of the applications of deep learning methods in trading. More complex models like Multilayer Perceptrons (MLP) and Long Short-Term Memory Networks (LSTM) can also model time dependencies and catch some of the higher-order statistics that were overlooked by simpler methods. Specifically:

LSTMs work well for financial time series since they maintain long-term dependencies, which allows them to discover lagging dependencies between paired instruments.

Convolutional neural networks (CNNs), used mainly for image processing, have also been used for extracting local trends and volatility patterns when used to model time series of structured data and this has shown promising results.

Autoencoders and Generative Adversarial Networks (GANs) also offer novel ways of denoising signals or generating synthetic pairs for training in situations when historical data is limited.

Reinforcement Learning (RL), are ideal for adaptive market strategies that require real-time responses to market changes, Deep Q-networks and Proximal Policy Optimization allow agents to learn optimal entry and exit rules by interacting with market environments.

2.3 Enhancing Model

Current Machine Learning models have varied and rich input features. Model performance can also be improved by incorporating other features such as sentiment scores from news/social media, macroeconomic indicators such as interest rates and inflation data and volatility indices like the VIX, as well as the historical OHLCV (open, high, low, close, volume) data. The growing utility of Natural Language Processing (NLP), according to Kaabar, is to extract those sentiment-based signals through alternative data. Van Der Post also emphasizes using alternative data in formulating methods for high-performing risk-adjusted returns.

2.4 Considerations and Challenges

The study on Comparative Analysis of LSTM, RNN, CNN and MLP For Stock Value Prediction by Takale et al. suggests that deep architectures capture much better the dynamics of price than shallow models. LSTM performs the best during volatile regimes due to its sensitivity to time. Furthermore, the Politecnico di Milano paper published in EJOR 2021 demonstrated the robustness of models across asset classes and highlighted the need for rolling window cross-validation to mitigate overfitting and assure generalization.

In addition, Kaabar discusses the drawbacks of single-period forecasting and presents multi-period modelling and transfer learning, both of which can be readily adapted to cross-asset strategies that involve equities, crypto, ETFs, and futures.

Practitioners can use platforms such as Zipline and PyFolio to set up trading environments to measure the model performance of Sharpe ratio, drawdown, turnover and PnL variance. Hands-On Financial Trading with Python stressed the need to include

slippage, commission and latency in backtesting environments when implementing ML-enhanced pair trading.

While ML/DL provide flexibility and adaptability, they pose challenges in time series modelling, including data leakage and overfitting.

The interpretability of deep models is still limited, although techniques like Shapley Additive Explanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) can give some insight.

Computational costs and the need for frequent retraining in dynamically changing markets can be quite significant.

Pythonic Quant, by Van Der Post, shared that even the most sophisticated models can be overwhelmed by rare events or black swan types of occurrences and structural breaks.

2.5 Research Gaps and Directions

The convergence of machine learning and long-short pair trading opens complementary avenues in the periphery of alpha discovery. Most of the empirical evidence showed that ML/DL-enhanced models outperformed the traditional ones, but their application needs much more careful attention to data engineering, feature selection, backtesting rigour and risk management. A gap still lies at the point where alternative data needs to be integrated with deep architecture for real-time deployment over multiple asset classes; this is a potentially promising direction for this research. Therefore, this research aims to fill that gap by integrating LSTM models specifically for real-time pair trading strategies in the digital asset domain to capture the dynamic, non-stationary behaviour of the ETH/BTC pair, offering a more adaptive, robust and profitable approach as compared to the traditional methods in this fast evolving and volatile asset class.

3. Competitor Analysis

The cryptocurrency market operates 24/7, causing high price fluctuations and generating a vast amount of data, which might lead to data inconsistencies or gaps in historical data (Jannatun & Ramya, 2024). The cryptocurrency market is also greatly influenced by various factors, including market sentiment and regulatory news (Kangsan & Daeyoung, 2023). This introduces noise and anomalies to the data, making it even more challenging to discover meaningful patterns.

The domain of cryptocurrency trading is increasingly characterized by the adoption of data-driven and algorithmic methodologies, reflecting the need for more sophisticated approaches in a highly volatile and complex market. In response to the limitations of traditional pairs trading strategies, this capstone project tackles the limitations of traditional pair trading by introducing an LSTM-based strategy, which is well-suited for analysing the time-series data to capture the long-term dependencies and complex, non-linear relationships. By employing an LSTM model, this capstone seeks to:

- Improve predictive accuracy: Generate more accurate predictions of the spread between ETH/USD and BTC/USD, leading to better-informed trading signals.
- Enhance adaptability: Dynamically adapt to the evolving market conditions and the non-stationary nature of cryptocurrency prices.
- Increase profitability: Achieve higher risk-adjusted returns compared to traditional pairs trading methods.

The potential impact of this project is significant. For traders and investors in the cryptocurrency market, this LSTM-based strategy offers a more robust and adaptive tool for navigating the volatility and complexity of ETH/BTC trading. It can lead to more efficient capital allocation, reduced risk, and increased profitability.

3.1 Possible Methodologies

Other competing methodologies have attempted to address the challenges posed by non-stationarity, regime shifts, and high volatility in asset relationships. To highlight these challenges, the following SWOT analysis of these key competitors summarises their respective strengths and weaknesses:

Table 1: Swot Analysis of Key Competitors:

Pair Trading Strategies	Strengths	Weaknesses
Traditional Statistical Arbitrage (cointegration, mean reversion and OLS)	• Low computational cost • Simple to implement and interpret • Widely used and academically grounded	• Assumption of stationarity and linearity • Poor in volatile market conditions
CNNs	• Effective for extracting local trends and volatility patterns from structured data	• Mainly designed for image processing • May miss long-term dependencies as compared with LSTMs
Autoencoders and GANs	• Good for denoising and anomaly detection • GANs can generate synthetic pairs for trading data augmentation	• Complex to train and tune • Interpretability and stability are limited
Ensemble models like XGBoost	• Effective for capturing non-linear relationships	• Require heavy feature engineering • Prone to overfitting if not carefully tuned
Reinforcement Learning	• Adaptive and responsive to real-time market changes	• Complex to implement • Require a lot of training data

4. Methodology and Approach

This research aims to develop a deep learning long/short trading strategy for ETH-USD and BTC-USD using a hybrid model GARCH-LSTM with attention to improved temporal feature extraction alongside an Altcoin Season Index (ASI) filter with real-time signal validation. GARCH naturally captures volatility but does not perform well in predictive accuracy for price movements, leveraging on integrating strength of GARCH and LSTM allows us to capture the statistical rigor and nonlinear pattern recognition. The ASI integration provides built-in risk controls by filtering signals during overextended price moves, complementing the statistical foundation of GARCH with technical validation. The method we employed in this research combines:

- GARCH for volatility clustering modelling
- LSTM for nonlinear pattern recognition in returns

- ASI Filter for enhancing trade timing

4.1 Returns

Price time series are the basis of the study. As listed in section 2, we will be using the prices of ETH-USD and BTC-USD, due to statistical reasons we will be using the returns of the series.

Returns can be calculated in two ways (Francisco, 2017), there are the discrete returns and the continuous compounded returns which we will be using in the research, although the returns of both are similar. The daily return r_t can be defined as

$$r_t = \log\left(\frac{p_t}{p_{t-1}}\right)$$

P_t is the price at time t and p_{t-1} being the price at time $t - 1$

4.2 GARCH model

The cryptocurrencies are known to be volatile, we will employ the GARCH model to predict the volatility of the Bitcoin and Ethereum returns separately, we will be applying the EGARCH which is defined as

$$\ln(h_t) = \omega + \beta \ln(h_{t-1}) + \alpha \left(\frac{\epsilon_{t-1}}{\sqrt{h_{t-1}}} \right) + \gamma \left(\left| \frac{\epsilon_{t-1}}{\sqrt{h_{t-1}}} \right| - E|z| \right)$$

where:

- h_t is the conditional variance
- $\omega, \alpha, \beta, \gamma$ are model parameters
- ϵ_t is the return innovation
- z is the standard normal random variable

The EGARCH model naturally accounts for the asymmetric impact of positive and negative returns on future volatility (leverage effect), which is highly prevalent in cryptocurrencies (Nelson, 1991; Andersen and Bollerslev, 1998).

4.2.1 Model Estimation and Diagnostics:

1. Fit EGARCH(1,1) model using maximum likelihood estimation (MLE)
2. Evaluate model fit through residual diagnostics (Box and Pierce, 1970), AIC/BIC criteria (Allen, 1971).
3. Compare EGARCH performance with realized volatility measures (Andersen et al, 2001)

4.3 LSTM - Attention Modeling

We will construct an LSTM network enhanced with Attention mechanism to predict the directional price movement.

4.3.1 LSTM Layer: Given an input sequence $X = (x_1, x_2, \dots, x_T)$ where x_t represents the feature vector at time t , the LSTM unit processes the input sequence in a forward and backward direction:

- Forward LSTM: This captures the sequence from $t = 1$ to $t = T$ to capture past-to-future dependencies.

$$\vec{h}_t = LSTM(x_t, \vec{h}_{t-1})$$

- Backward LSTM: This on the other hand processes sequence from $t = T$ to $t = 1$ to learn future-to-past context.

$$\overleftarrow{h}_t = LSTM(x_t, \overleftarrow{h}_{t+1})$$

- Final hidden state: Finally concatenated the hidden states to preserve both directional context.

$$h_t = \begin{bmatrix} h_t^{\rightarrow} \otimes h_t^{\leftarrow} \end{bmatrix}$$

where:

- h_t is the hidden state
- $\otimes$ is concatenation

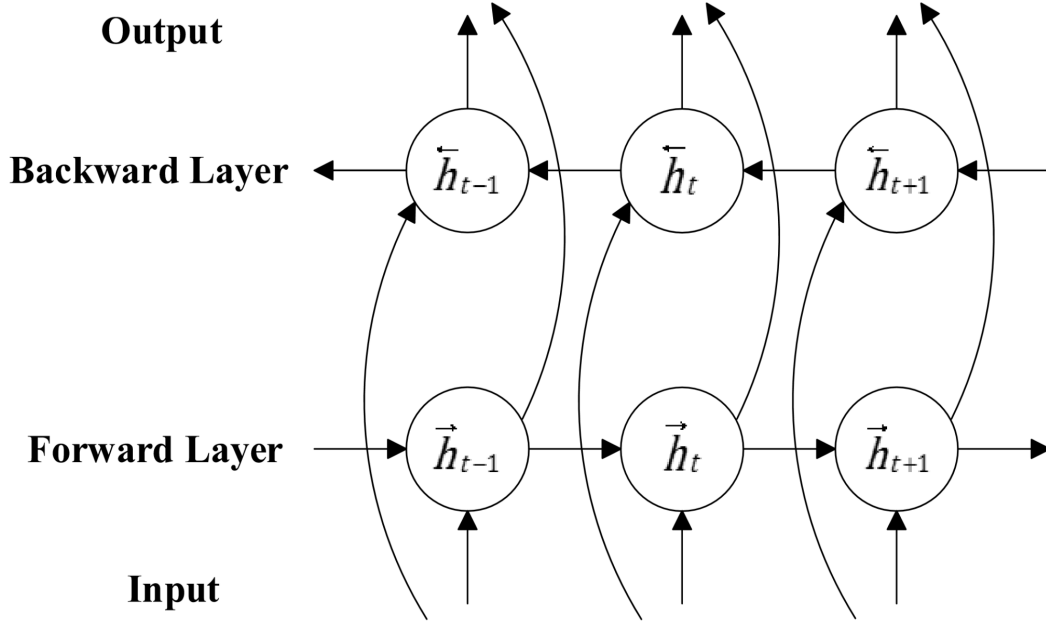


Figure 1: LSTM model

4.3.2 Multi-Head Attention Layer: At this stage the hidden states h_t are split into queries (Q), keys(K), and values(V), the attention score is then computed to get the similarity between Q and K to weight V. Finally we proceed to combine outputs from N parallel attention heads in a process called Multi-Head Fusion.

$$Q = hW^Q, K = hW^K, V = hW^V$$

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

$$MultiHead(h) = [head_1 \otimes \dots \otimes head_N]W^O$$

where

Q is the Query matrix (learns “what to look for”)

K is the key matrix (learns “relevance to query”)

V is the value matrix (learns “feature importance”)

$d_k = d_v = \frac{d_{model}}{B}$ which is the dimension per head

N is the number of attention heads typically 4-8

With this implementation the model will be able to identify salient features (such as sudden spikes) and the parallel heads will learn diverse temporal patterns.

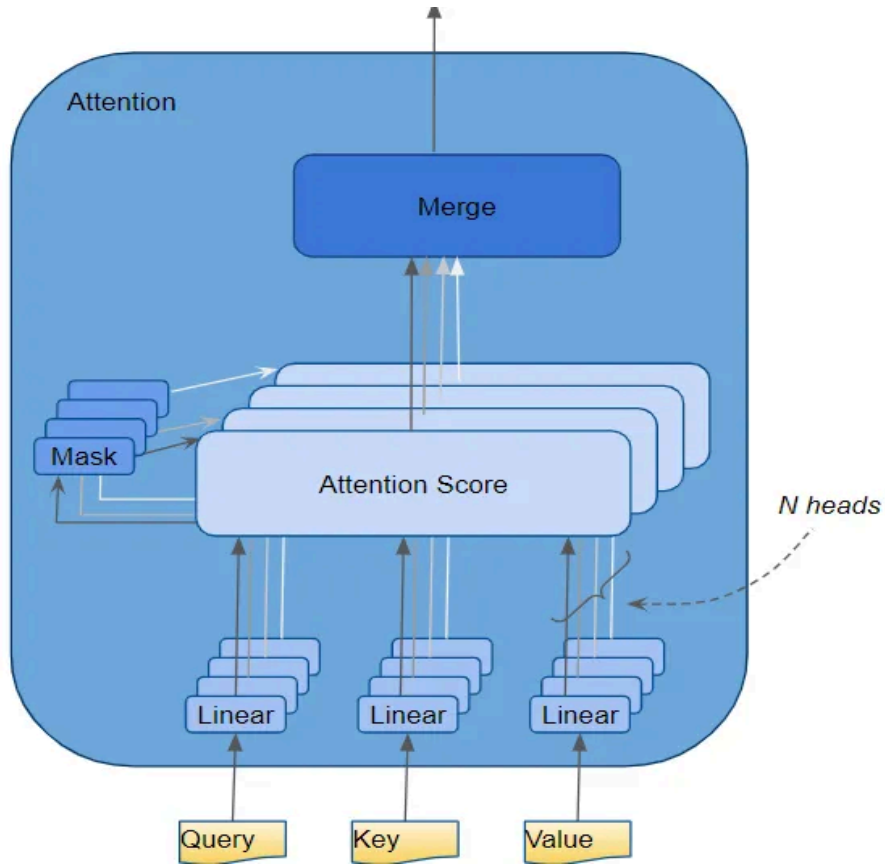


Figure 2: Multi-Head Attention

4.3.3 Temporal Attention Layer: This is a dynamic time selection mechanism when paying attention (Xu et al, 2017), it assigns dynamic weights to time steps based on relevance.

$$score(h_t) = v^T \tanh(Wh_t + b)$$

$$\alpha_t = \frac{\exp(score(h_t))}{\sum_t \exp(score(h_t))}$$

$$c = \sum_t \alpha_t h_t$$

where

α is the attention weight or time step t

c is the context vector (weight sum of hidden states)

W is the weight matrix

v is the weight vector

b is the Bias term

4.3.4 Output Layer: The output layer generates profitable trading signals and volatility forecasts are the result and are given by

$$\hat{y}_t = \text{softmax}(W_y c + b_y)$$

$$\hat{\sigma}_t = \text{ReLU}(W_\sigma c + b_\sigma)$$

where

$\hat{y}_t$ is the Predicted probabilities for {Long, Short, Neutral}

$\hat{\sigma}_t$ is the Forecasted Volatility

W_y is the classification weights
 b_y is the classification bias
 W_σ is the Volatility regression weights
 b_σ is the Volatility bias

4.4 ASI Filter

The ASI filter implements real-time signal validation through technical momentum thresholds, enhancing the LSTM's directional predictions with volatility-aware position control. A 30-day simple moving average of the ETH/BTC spread identifies the market regimes:

$$MA_t^{30} = \frac{1}{30} \sum_{i=t-29}^t S_i$$

where

S_i is the ETH/BTC spread value at time i

This formula computes the arithmetic mean of the most recent 30 spread values, providing a smoothed trend indicator that filters out short-term price noise while preserving longer-term directional movements in the ETH/BTC relative performance.

4.4.1 Regime Classification: The ASI is determined by comparing the current ETH/BTC spread to its past trend:

$$Regime_t = \begin{cases} 1 & \text{if } S_t > MA_t^{30} \\ 0 & \text{otherwise} \end{cases}$$

This creates a binary flag that it is altcoin season if the spread is greater than the 30 day moving average. The 30-day window balances responsiveness to trend changes while smoothing short-term volatility.

5. Results

We obtained BTC/USD and ETH/USD daily closing prices using Alpaca's API. The log return time series plot below shows that BTC and ETH fluctuate a lot, implying high volatility. However, this volatility decreases gradually over time with smaller fluctuations.

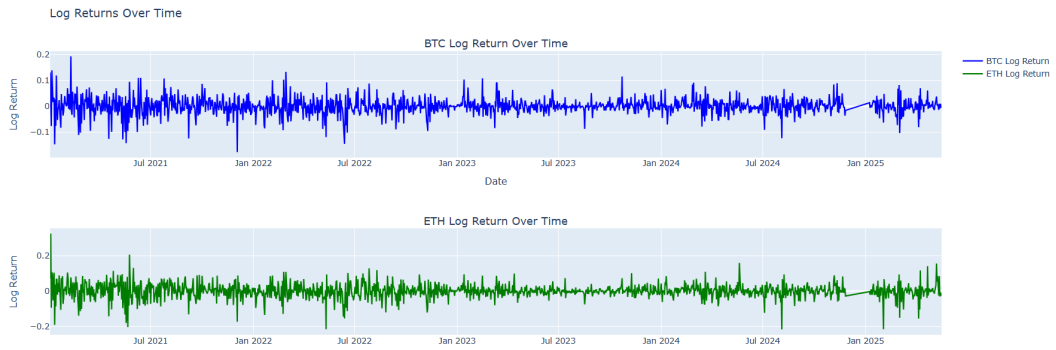


Figure 3: BTC and ETH Log Return

5.1 GARCH Model

We employ the EGARCH (1,1) model to the BTC and ETH return using maximum likelihood estimation (MLE). The model performance is also evaluated through residual diagnostics using the Ljung-box test. Below is the model performance on respective train data and test data:

5.1.1 BTC/USD

Model Fit. The test data achieved a higher log-likelihood and lower AIC/BIC, indicating that the test data generalises well without overfitting.

Table 2: EGARCH Model Fit Performance for BTC/USD

	Train Data	Test Data
log-likelihood	-3232.24	-743.74
AIC	6470.47	1493.47
BIC	6485.9	1504.74

Parameters' coefficient. The coefficients of omega and alpha in train data are not significant, while beta is significant. While all three parameters of the test set are significant, indicating that the model can better interpret volatility dynamics.

Both datasets have high beta values closer to 1, reflecting that the volatility is highly persistent over time.

Table 3: EGARCH Model Fit Performance for BTC/USD

	Train Data	Test Data
omega	0.0187 (p-value: 0.438)	0.3167 (p-value <0.05)
alpha	0.0333 (p-value: 0.466)	0.2965 (p-value <0.05)
beta	0.9926 (p-value <0.05)	0.8428 (p-value <0.05)

Residual Diagnostics. Both datasets' test p-values are above 0.05, indicating no significant autocorrelation in residuals. The model has successfully captured the underlying volatility pattern, especially test data has outperformed the train data in this aspect with high p-value at 0.85, implying good generalisation.

Table 4: EGARCH Model Fit Performance for BTC/USD

	Train Data	Test Data
Ljung-Box test p-value	0.0733	0.8528

The plots below visualise the EGARCH performance with realised volatility measures.

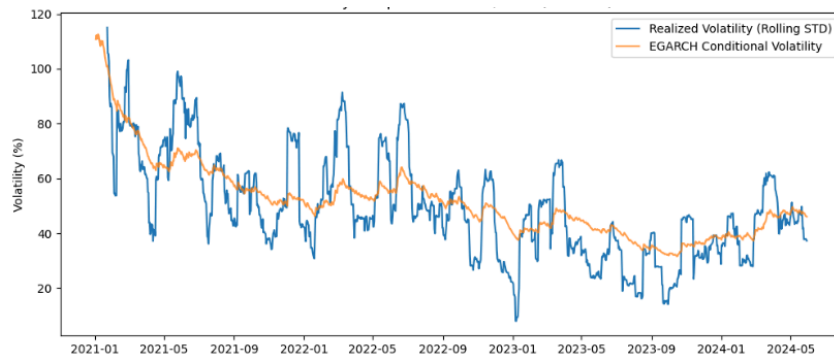


Figure 4: Volatility Comparison for BTC/USD (Train Set)

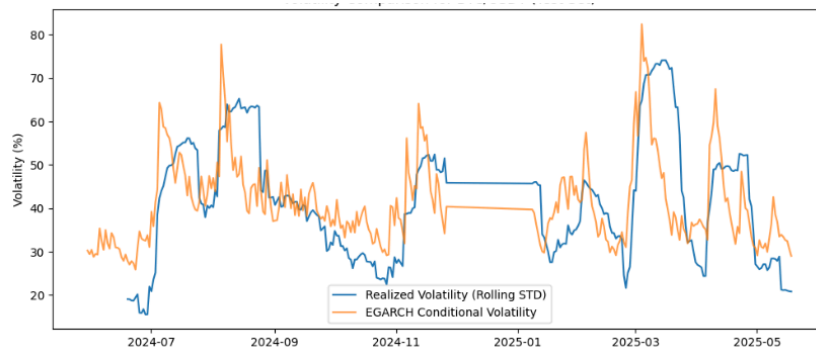


Figure 5: Volatility Comparison for BTC/USD (Test Set)

5.1.2 ETH/USD

Model Fit. The test data has a higher log-likelihood and lower AIC/BIC, indicating that the test data generalises well without overfitting.

Table 5: EGARCH Model Fit Performance for ETH/USD

	Train Data	Test Data
log-likelihood	-3462.45	-877.84
AIC	6930.90	1761.69
BIC	6946.32	1772.96

Parameters' coefficient. Omega and alpha values of both datasets are significant, indicating that the models have captured the dynamics. However, the beta of the test dataset is not significant, implying that the test dataset model might not capture the effect consistently.

Table 6: EGARCH Model Fit Performance for ETH/USD

	Train Data	Test Data
omega	0.0555 (p-value <0.05)	1.7103 (p-value <0.05)
alpha	0.1194 (p-value<0.05)	0.6246 (p-value <0.05)
beta	0.9813 (p-value <0.05)	0.3998 (p-value: 0.108)

Residual Diagnostics. Both datasets' test p-values are also above 0.05, indicating no significant autocorrelation in the residual.

Table 7: EGARCH Model Fit Performance for ETH/USD

	Train Data	Test Data
Ljung-Box test p-value	0.2488	0.9938

The plots below visualise the EGARCH performance with realised volatility measures.

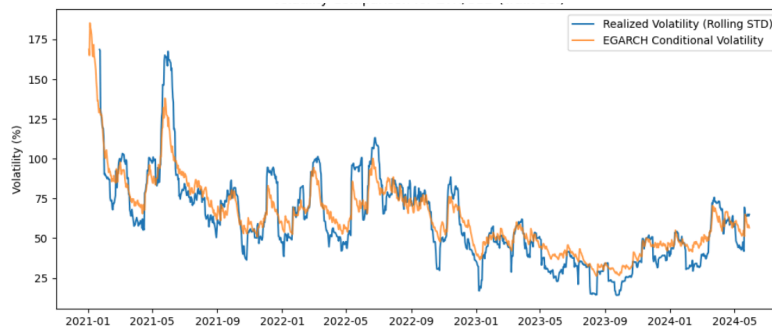


Figure 6: Volatility Comparison for ETH/USD (Train Set)

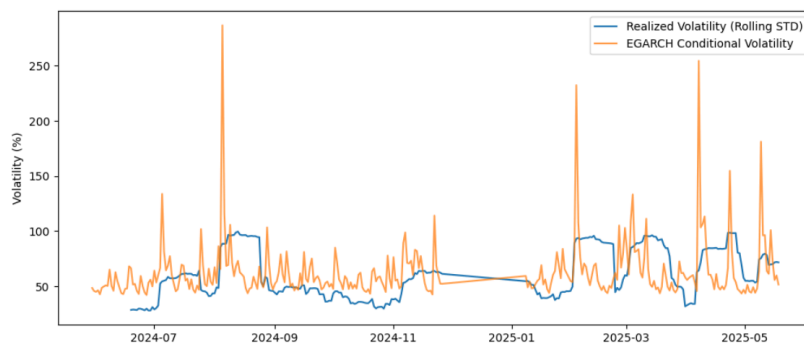


Figure 7: Volatility Comparison for ETH/USD (Test Set)

Overall, both test datasets fit the model better than the train datasets. This could be attributed to external factors such as changes in market dynamics and regime. For instance, the crypto market in the 2021 - 2024 period is more volatile with more abrupt events like spikes during the COVID period, compared to a slightly more stable market in 2024. Besides, due to the limitation of historical data obtained from Alpaca API, the

duration of datasets is shorter, which could lead to better performance due to less complexity.

5.2 LSTM Model

The model was trained using spread return and EGARCH volatility forecasts.

Training set accuracy ranges from 76% to 85%, while validation set accuracy fluctuates between 74% to 86%. It is observed that both accuracy has been steadily improving and performing well, although they have a ceiling at around 86.

However, the fluctuation is likely due to limited data, indicating that there's room to improve the signal accuracy.

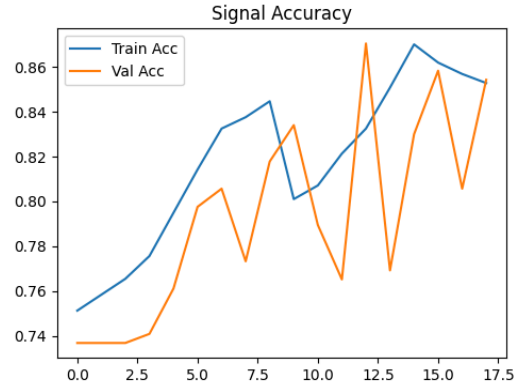


Figure 8: Signal Accuracy

Both curves of the training set and validation set are aligned, suggesting that there's no overfitting. However, the MSE of the training set dropped sharply from 0.3 to around 0.005 before stabilising. The validation set has maintained a low and stable MSE throughout.

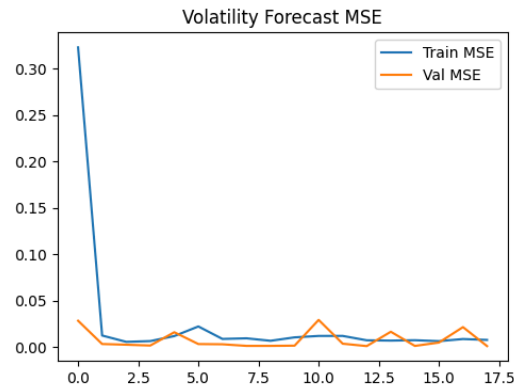


Figure 9: Volatility Forecast MSE

5.3 Backtesting

Prior to conducting the backtest, we implemented an ASI as a critical signal validation layer. This technical indicator is integrated into our GARCH-LSTM framework to enhance decision-making robustness and mitigate exposure to overextended market conditions. The ASI filter operates as a conditional gate for our model's prediction. It will only execute long positions on ETH/BTC when ASI is greater than MA_t^{30} which indicates the altcoin season, vice versa, short positions will only be executed when ASI is lesser

and equal to MA_t^{30} to avoid counter-trend exposure during Bitcoin dominance. This approach synergises the model's pattern recognition capabilities with traditional momentum analysis.

The strategy's performance without the ASI filter is plotted as below:

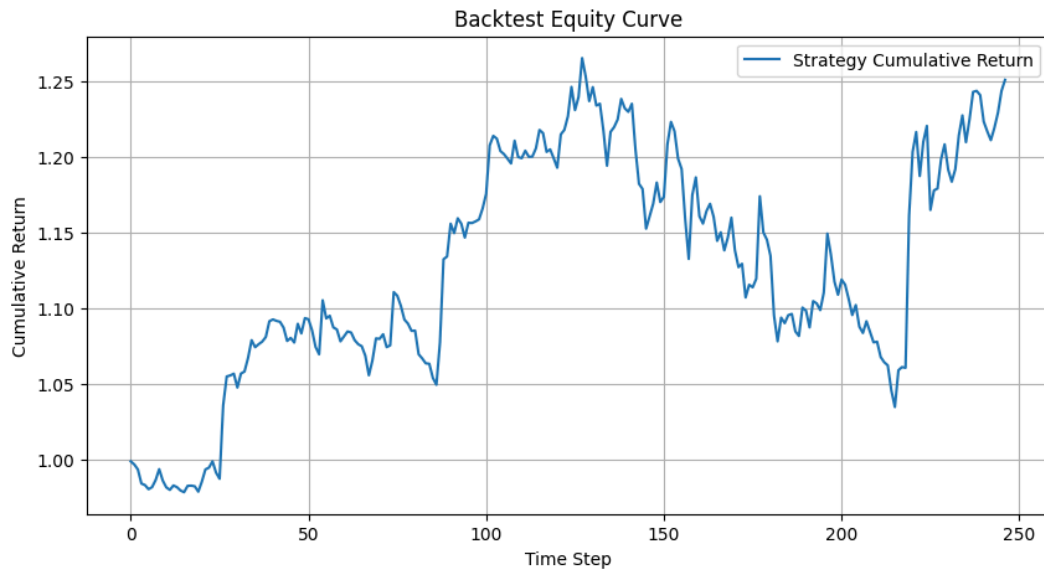


Figure 10: Backtest Equity Curve within the Test Period

After applying the ASI filter, the strategy's performance is improved gradually as below:

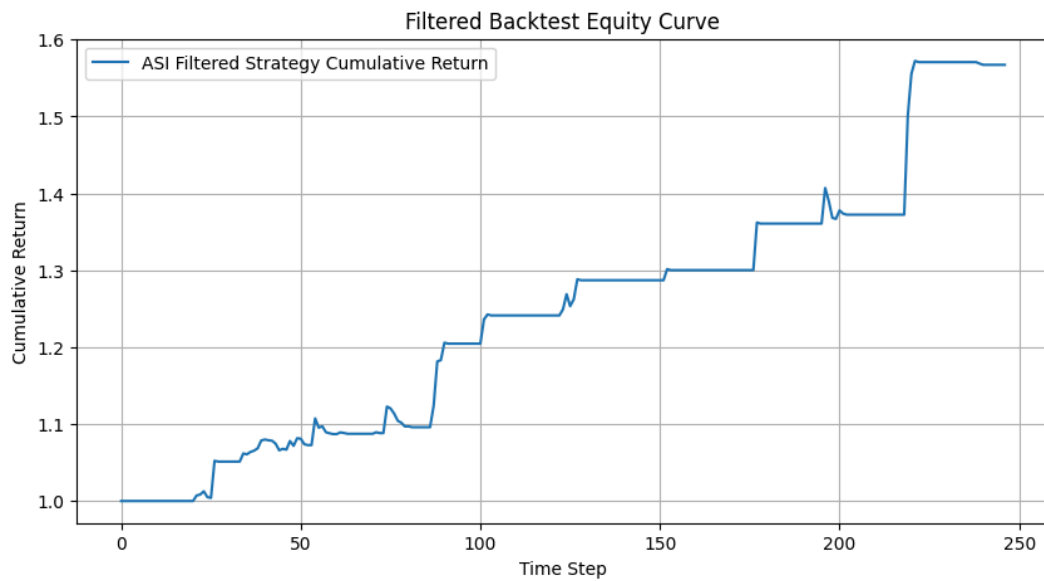


Figure 11: Filtered Backtest Equity Curve within the Test Period

The strategy demonstrated robust performance across key metrics:

Table 8: Backtest Performance of GARCH-LSTM with ASI filter

Metric	Value
Total Return	156.71%
Sharpe Ratio	3.01
Maximum Drawdown	-4.02%

Table 9: Comparison between GARCH-LSTM with and without ASI filter

Metric	Unfiltered Strategy	ASI-Filtered Strategy
Total Return	125.09%	156.71%
Sharpe Ratio	1.14	3.01
Maximum Drawdown	-23.03%	-4.02%

6. Discussion

The findings of this capstone demonstrate the effectiveness of integrating traditional econometric models with modern machine learning techniques for ETH/BTC pair trading. The addition of an ASI filter to our hybrid GARCH-LSTM model marks a significant improvement over conventional statistical arbitrage, providing a much-needed edge in the notoriously volatile crypto market.

7. Conclusion

In order to address the issues of volatility clustering and non-linear price dynamics associated with trading ETH/BTC pairs, the study proposes a hybrid model that combines GARCH and LSTM with attention alongside an ASI filter. Three key components make up the model's architecture:

EGARCH(1,1) demonstrated a good statistical fit by AIC/BIC criteria and captures asymmetric effects like the leverage effect seen in cryptocurrency markets by modelling volatility. Additionally, it passed the independence residual diagnostic tests (Ljung-Box: $p > 0.05$), demonstrating its suitability for simulating long-term volatility for both Bitcoin and Ethereum.

In the classification of trading signals (long, short, and neutral), LSTM with Multi-Head Attention achieved a validation set accuracy of 86% in predicting directional changes in spread. By enabling the network to concentrate on significant temporal features, the Attention mechanism improved predictions and made the model more interpretable.

ASI filter improves trade execution by preventing short position holding during altcoin season. Higher returns and more stability were the results of this extra restriction. The ASI-filtered approach has performed better in terms of risk-adjusted returns as well as absolute returns. This model's outstanding risk-return efficiency is demonstrated by its Sharpe ratio of 3.01, and better capital preservation is indicated by a maximum drawdown of negative 4.02%

The hybrid GARCH-LSTM-Attention model is a perfect example of how to integrate traditional econometrics with deep learning into quantitative trading. This model is able to model volatility dynamics, extract features from sequential data and by implementing an ASI filtering on execution, enabling a robust cryptocurrencies pair trading.

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Appendix

GitHub Repository:

<https://github.com/zheyuzy/GARCH-LSTM-model-on-ETH-BTC-pair>